Midterm 2 — Sample 2

DATA 412/612

Practice exam, questions only. Weeks 1–11. The material since Midterm 1 is joins, strings, factors, dates, and statistics (a nested model comparison and numerical summaries).

Put your answers in an R Markdown document and keep the code visible (`knitr::opts_chunk$set(echo = TRUE)` in the setup chunk). You may use your notes and the course website. You may not use Google, any other website, or an LLM. Do not discuss the exam with anyone else. Load the tidyverse. `mpg` comes with `ggplot2`.

1. **Recode a factor** (8 pts). In `mpg`, collapse `drv` so that `"f"` and `"r"` become `"two"` and `"4"` stays `"four"`. Drop unused levels. Report how many cars are in each new level.

2. **Periods and zones** (8 pts). `start` is a date-time in `America/New_York`. Build the instant two weeks and three hours later, using a period. Also report the length of the interval from `start` to that later instant in hours. Show `start` again in `UTC`.

```
start <- ymd_hm("2024-11-03 01:15", tz = "America/New_York")
```

3. **Nested models** (10 pts). Fit two models on `mpg`: `hwy` on `class` alone, and `hwy` on `class` plus `displ`. Compare them with `anova()`. In one sentence, say whether `displ` improves the fit after `class` is already in the model. Use the p-value from that comparison.

4. **Summaries** (8 pts). For each `class` in `mpg`, report the number of cars, the median `hwy`, and the interquartile range of `hwy`. Sort by the median, highest first.

5. **Strings, then a date** (6 pts). The vector below uses a dot between the pieces and a trailing letter that is not part of the date. Remove that letter and parse what remains as a date. Print the dates.

```
c("2024.03.15x", "2023.12.01x", "2022.07.04x")
```